

STUDY REPORT

Study Title: Validation of an ELISA Method for the
Quantification of LT3114 in a Citrate Buffer
Formulation

Study Number: BRT 20140629-A

Total Pages 50

Testing Facility: Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560, USA

Report Date: 21 November 2014



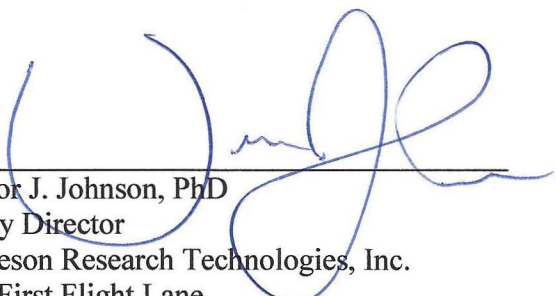
STUDY IDENTIFICATION

Study Title	Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation	
Study Number	BRT 20140629-A	
Sponsor	Lpath Incorporated 4025 Sorrento Valley Boulevard San Diego, CA 92121	
Sponsor Representative	Gary Woodnutt, PhD (Lpath)	
Study Monitor	Marlene W. Modi, PhD (Consultant for Lpath) MWM Consulting Group, Inc.	
Testing Facility	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560	
Study Director	Victor J. Johnson, PhD (BRT)	
Study Schedule	Study Initiation:	01 August 2014
	Experimental Start:	05 August 2014
	Experimental Completion:	13 October 2014
	Study Completion:	Date Study Director signed the final report.

COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)
- OECD Principles on Good Laboratory Practice (C[97]186/Final)



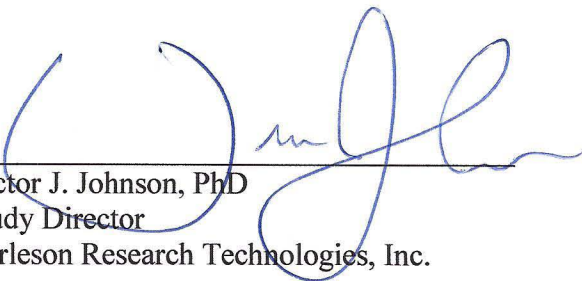
Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
USA

21 Nov 14

Date

APPROVALS

Study Director Approval

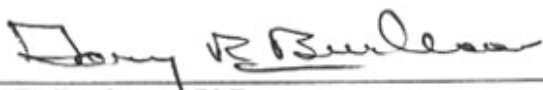


Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.

21 Nov 14

Date

BRT Management Approval



Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

21 NOV 2014

Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

21 Nov 2014

Date

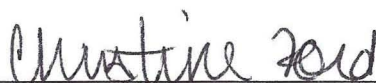
QUALITY ASSURANCE STATEMENT

Study Number: BRT 20140629-A
Study Title: Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation
Sponsor: Lpath Incorporated
Testing Facility: Burleson Research Technologies, Inc.
Study Director: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to Study Director/Management
Protocol Review	01 Aug 14	01 Aug 14
Quality Controls Preparation Audit	13 Aug 14	13 Aug 14
Sample Addition Audit	15 Aug 14	15 Aug 14
Amendment 1 Review	26 Aug 14	26 Aug 14
Raw Data 1 Audit	14 Oct 14	14 Oct 14
Raw Data 2 Audit	21 Oct 14	21 Oct 14
Report Audit	24 Oct 14	24 Oct 14
Report 2 Audit	29 Oct 14	29 Oct 14
Amendment 2 Review	13 Nov 14	13 Nov 14
Amendment 3 Review	18 Nov 14	18 Nov 14
Final Report Audit	19 Nov 14	19 Nov 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.



Christine Ford
Quality Assurance Manager



Date

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SUMMARY

The present study was conducted to validate an ELISA method for the quantification of LT3114 in a citrate buffer formulation (Lpathomab-DS). Validation parameters that were examined included determination of limit of detection (LOD), citrate buffer interference, calibration curve performance, range of quantification, accuracy, precision, linearity, stability evaluation, and robustness/ruggedness. The results demonstrate the validation goals were achieved and the assay is suitable for the purpose of detecting LT3114 in a citrate buffer formulation.

The limit of detection was determined by analyzing multiple blank wells containing delipidized bovine serum (DBS) diluent across multiple assays. The LOD was calculated at 0.454 ng/mL. This LOD value is lower than the limit of quantification of 3.9 ng/mL of LT3114. This validation determined the quantifiable range of the method to be 3.9 – 500 ng/mL LT3114. The citrate formulation buffer did not interfere with assay signal as demonstrated by lack of signal for all dilutions tested, including neat.

LT3114 in a citrate buffer formulation was used to demonstrate linearity of dilution of the ELISA method. Acceptable dilution linearity was demonstrated by acceptable accuracy and precision of dilutions from 1:80,000 to 1:1,280,000. This parameter demonstrated the ability to dilute samples with a high concentration of LT3114 into the range of the calibration curve and provided acceptable point estimates for the concentration of LT3114 in dose formulation samples.

QC samples were prepared by spiking Human Reference anti-LPA IgG LT3114 into citrate formulation buffer and were used to determine the reproducibility (accuracy and precision) of the ELISA assay. Intra- and inter-assay accuracy and precision were acceptable. These parameters were used to measure the stability of LT3114 under a variety of conditions and in establishing acceptance criteria for the validated method.

The QCs were stable for ≤ 4 freeze/thaw cycles and following storage at RT for 4 hours. The QCs were stable when stored frozen ($\leq -70^{\circ}\text{C}$) for up to 61 days and the LT3114 in a citrate buffer formulation was stable when stored at 2-8 $^{\circ}\text{C}$ for up to 99 days. Additional analyses of long-term stability, if required by the Sponsor, may be performed and added as an addendum to this report.

In addition, assay robustness was demonstrated using different plate readers and two independent operators.

The validation indicated that this method is suitable for quantification of LT3114 in dose formulation samples prepared with citrate formulation buffer.

ABBREVIATIONS

BSA	-	Bovine Serum Albumin
%CV	-	% Coefficient of Variation
DBS	-	Delipidized Bovine Serum
GLP	-	Good Laboratory Practice
HRP	-	Horse Radish Peroxidase
IgG	-	Immunoglobulin G
LLOQ	-	Lower Limit of Quantification
LOD	-	Limit of Detection
LPA	-	Lysophosphatidic Acid
OD	-	Optical Density
QC	-	Quality Control
%RE	-	% Relative Error
TMB	-	3,3',5,5'-Tetramethylbenzidine

PURPOSE

The purpose of this study was to validate an ELISA method for quantification of LT3114 prepared in citrate buffer. The validated assay will be used for dose formulation analysis on selected studies conducted by the Sponsor. This assay utilized lysophosphatidic acid (LPA) conjugated to bovine serum albumin (BSA) as the capture reagent. LT3114 formulated in citrate buffer was applied to a solid substrate coated with LPA-BSA followed by detection of captured LT3114 using peroxidase-conjugated anti-human IgG.

Assay performance parameters evaluated during validation included: limit of detection (LOD), citrate buffer interference, calibration curve performance, range of quantification, accuracy, precision, linearity, stability evaluation, and robustness/ruggedness.

MATERIALS

LT3114 in a citrate buffer formulation: Lpathomab-DS, 30 mg/mL

Description:	Human IgG monoclonal antibody specific for lysophosphatidic acid (LPA) formulated in citrate formulation buffer
Manufacturer:	Gallus Biopharmaceuticals
Batch/Lot Number:	Gallus-MO 103948 (see Appendix I for Certificate of Analysis)
Concentration:	32.1 mg/mL
Receipt Date:	01 Jul 14
Expiration Date:	Undergoing stability assessment by Sponsor; Retest date is Dec 2014.
Storage Conditions:	2-8°C

Citrate Formulation Buffer

Description:	24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate Dibasic, 300 mM D-Trehalose, pH 5.0
Manufacturer:	Gallus Biopharmaceuticals
Batch/Lot Number:	104011-001 (see Appendix II for Certificate of Analysis)
Receipt Date:	01 Jul 14
Expiration Date:	Mar 2015
Storage Conditions:	2-8°C

Human Reference Anti-LPA IgG LT3114

Description:	Human IgG monoclonal antibody specific for lysophosphatidic acid (LPA)
Manufacturer/Supplier:	Lpath Incorporated
Batch/Lot Number:	LP121416 and LP121430 (see Appendix III for Certificate of Analysis)
Concentration:	20.1 mg/mL
Receipt Date:	21 May 14
Expiration Date:	Undergoing stability assessment by Sponsor
Storage Conditions:	Stored at $\leq -70^{\circ}\text{C}$ until use. Stable for up to 6 months at 2-8°C once thawed per Sponsor. (see Appendix IV)

Capture Material: LPA-BSA

Description:	12OLPA-SMCC-BSA [LPA conjugated to bovine serum albumin (BSA)]
Manufacturer/Supplier:	Lpath Incorporated
Batch/Lot Number:	LP121430 (see Appendix V for Certificate of Analysis)
Concentration:	5 mg/mL
Receipt Date:	21 Aug 14 and 05 Jun 14
Storage Conditions:	Stored at $\leq -70^{\circ}\text{C}$ until use. Stable for up to 6 months at 2-8°C once thawed per Sponsor. (see Appendix VI)

METHOD

The ELISA method used for validation of LT3114 in a citrate buffer formulation is presented in [Appendix VIII](#).

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. LT3114 in a citrate buffer formulation, standards, quality control (QC) samples, and dilipidized bovine serum (DBS) assay buffer were added to the coated plates and incubated to allow for capture of the LT3114. A horse radish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Data Analysis

Data analysis was performed using the SoftMax®Pro software v 2.2.1 (Spectramax 340) and v 5.0.1 (Spectramax M2e). A standard curve was generated by plotting the resulting mean absorbance at 450 nm (OD₄₅₀) vs. concentration and fitting the curve using a 4 parameter logistics (PL) equation. Printouts included absorbance values (raw data), the standard curve(s), the coefficient of variation between replicates, and the interpolated value of the unknown(s). Results were expressed using two decimals in accordance with the decimals used for the calibration curve standards.

Statistical Analysis

No statistical analysis of the data was performed except for calculation of means, standard deviations, %RE (relative error as a measure of accuracy), and %CV (coefficient of variation as a measure of precision), as appropriate. These data calculations were performed using SoftMax®Pro (version 2.2.1 and 5.0.1 as appropriate) and/or Microsoft Excel 2013.

VALIDATION PARAMETERS

The validation of the ELISA for determination of LT3114 in a citrate buffer formulation included determination of LOD, citrate buffer interference, calibration curve performance, range of quantification, accuracy, precision, linearity, stability evaluation, and robustness/ruggedness. A summary table of all assay runs is provided in [Appendix VII](#).

LT3114 was provided to the Testing Facility as a formulation prepared in citrate buffer at 32.1 mg/mL. The dose levels that the Sponsor has selected to test in GLP studies are 30 mg/mL, 10 mg/mL, and 2 mg/mL of LT3114 in a citrate buffer formulation. Therefore, the assay was validated to determine LT3114 concentrations in a citrate buffer formulation. In addition, the limit of detection was determined to ensure that the assay is suitable to confirm the absence of LT3114 in vehicle preparations.

Limit of Detection

The LOD established during the method transfer (BRT 20140420) was confirmed by assessing the background signal from wells receiving only DBS performed in 4 assays each with ≥ 10 blanks (see [LOD results section](#) for details). The LOD was calculated as follows:

$$\text{LOD} = \text{Mean}_{\text{blanks}} + 2\text{SD}_{\text{blanks}}$$

where SD=standard deviation.

Citrate Buffer Interference and LT3114 Recovery

Citrate buffer (provided by the Sponsor) was tested for signal in the assay by testing a dilution series of citrate buffer (neat to 1:1000) in 4 assays. The same dilution series of citrate buffer was spiked with 5 ng/mL and 10 ng/mL of Human Reference LT3114 (Anti-LPA IgG antibody) to determine recovery from citrate buffer at low levels of LT3114. A control spike was prepared in delipidized bovine serum (DBS) assay diluent near the lower limit of quantification (LLOQ) at 5 ng/mL and 10 ng/mL.

Samples were evaluated for acceptable recovery of $\pm 25\%$ of nominal.

Calibration Curve and Range of Quantification

The calibration curve (standard curve) parameters were evaluated from 16 calibration curves. The back calculated concentration of the standards and the mean accuracy values were calculated. The range of quantification was determined by assessment of accuracy of the back calculated values for each standard.

Acceptable performance of the calibration curves was defined as calibration standards within $\pm 20\%$ of the nominal value [25% at the upper limit of quantification (ULOQ) and LLOQ]. The range of quantification was established as the highest and lowest concentrations that had an accuracy within 25% of the nominal concentration.

Accuracy and Precision

The accuracy of the method was determined by evaluating the closeness of the mean result to the nominal value of replicate analysis of 3 quality control (QC) concentrations (QC1 – 500 ng/mL, QC2 – 150 ng/mL, and QC3 – 5 ng/mL). The precision of the method was determined by evaluating the closeness of individual measures by analysis of 3 QC concentrations in duplicates. Acceptable within-run and between-run accuracy was within 20% of nominal for each QC (25% at LLOQ and ULOQ). Acceptable within-run and between-run precision did not exceed a CV% of 20% for each QC (25% at LLOQ and ULOQ). Acceptable total error [sum of absolute value of accuracy expressed as % relative error (%RE) and precision] for each QC was less than 30% (40% at LLOQ and ULOQ).

- Intra-assay accuracy was assessed from 6 analyses of each of 3 QC samples in 1 (QC1 – 500 ng/mL) or 2 (QC2 – 150 ng/mL and QC3 – 5 ng/mL) assay runs.
- Inter-assay accuracy was assessed from a single analysis of the 3 QC samples in 7 (QC1 – 500 ng/mL) or 13 (QC2 – 150 ng/mL and QC3 – 5 ng/mL) assay runs.

Linearity of Dilution

Dilution integrity was demonstrated by determining the LT3114 concentration of 6 serial 2-fold dilutions of the LT3114 in a citrate buffer formulation (provided by the Sponsor) in ≥ 5 replicates in 2 assays. Accuracy and precision were calculated for each dilution and acceptable results did not exceed 20%, and the total error (sum of

absolute value of accuracy and precision) for each acceptable dilution was less than 30% (40% at LLOQ and ULOQ).

Stability

Stability determination was performed using aliquots of the high (QC1 – 500 ng/mL) and low (QC3 – 5 ng/mL) QC tested against a calibration curve of freshly prepared standards. Acceptable stability was achieved when accuracy was within 20% of the nominal value.

- Benchtop stability of High and Low QC at room temperature for ≥ 4 hours.
- Freeze-thaw stability of High and Low QC was assessed for up to 4 freeze-thaws. For freeze-thaw stability, frozen samples were thawed at room temperature and refrozen for at least 12 hours prior to each subsequent thaw.
- Stability of High and Low QC and at $\leq -70^{\circ}\text{C}$ for ≤ 4 weeks.
- Stability of the LT3114 in a citrate buffer formulation at $2-8^{\circ}\text{C}$ for ≤ 4 weeks.
- Long term stability of the High and Low QC at $\leq -70^{\circ}\text{C}$ and the LT3114 in a citrate buffer formulation may be assessed for ≤ 3 months. If additional long term stability is required and the samples are available, the results of additional analyses may be reported in this validation report or by an addendum to the validation report.

Robustness/Ruggedness

The following parameters were evaluated during validation for robustness:

- different plate readers
- different operators

Two plate readers (Spectramax 340 and M2e) were used to read the same plate for two independent assays. Acceptable accuracy and precision of the standard curve and QCs was used to confirm robustness of the assay to multiple plate readers.

Assay performance was compared for two independent operators. Each operator performed at least 1 assay.

RESULTS

Limit of Detection

The LOD was calculated based on the mean plus 2 standard deviations of the assay signal produced by blank wells containing only DBS (Table 1). The LOD was determined to be an assay signal OD₄₅₀ of 0.058. The absorbance value for the LOD was converted to the mean LT3114 concentration of 1.48 ng/mL by interpolating the OD₄₅₀ value from the calibration curves from all 4 assays performed for LOD assessment.

Table 1: Assay signal from blank wells containing DBS diluent.

Assay Run	Blank (OD ₄₅₀)	Assay Run	Blank (OD ₄₅₀)
08Aug14	0.046	14Aug14	0.042
08Aug14	0.045	14Aug14	0.046
08Aug14	0.044	15Aug14 (Operator 1)	0.060
08Aug14	0.046	15Aug14 (Operator 1)	0.056
08Aug14	0.049	15Aug14 (Operator 1)	0.050
08Aug14	0.046	15Aug14 (Operator 1)	0.048
08Aug14	0.044	15Aug14 (Operator 1)	0.051
08Aug14	0.046	15Aug14 (Operator 1)	0.053
08Aug14	0.044	15Aug14 (Operator 1)	0.058
08Aug14	0.046	15Aug14 (Operator 1)	0.055
08Aug14	0.044	15Aug14 (Operator 1)	0.052
08Aug14	0.046	15Aug14 (Operator 1)	0.051
08Aug14	0.058	15Aug14 (Operator 1)	0.058
08Aug14	0.051	15Aug14 (Operator 1)	0.050
08Aug14	0.054	15Aug14 (Operator 1)	0.053
08Aug14	0.047	15Aug14 (Operator 1)	0.059
08Aug14	0.051	15Aug14 (Operator 1)	0.060
08Aug14	0.051	15Aug14 (Operator 1)	0.060
08Aug14	0.049	15Aug14 (Operator 1)	0.056
08Aug14	0.049	15Aug14 (Operator 1)	0.056
08Aug14	0.048	15Aug14 (Operator 2)	0.047
08Aug14	0.053	15Aug14 (Operator 2)	0.043
14Aug14	0.047	15Aug14 (Operator 2)	0.042
14Aug14	0.042	15Aug14 (Operator 2)	0.042
14Aug14	0.047	15Aug14 (Operator 2)	0.046
14Aug14	0.045	15Aug14 (Operator 2)	0.043
14Aug14	0.045	15Aug14 (Operator 2)	0.046
14Aug14	0.042	15Aug14 (Operator 2)	0.043
14Aug14	0.039	15Aug14 (Operator 2)	0.044
14Aug14	0.044	15Aug14 (Operator 2)	0.042
14Aug14	0.042	15Aug14 (Operator 2)	0.042
14Aug14	0.048	15Aug14 (Operator 2)	0.049
		Mean	0.048
		SD	0.005
		%CV	10.4
		LOD (OD ₄₅₀)	0.058
		LOD (ng/mL)	1.48

Citrate Buffer Interference and Recovery of LT3114

The citrate formulation buffer was tested for signal in the assay from neat to 1:1000 dilution in DBS. As shown in [Table 2](#), citrate buffer did not result in detectable assay signal.

Table 2: Background assay signal from citrate formulation buffer.

Citrate Buffer Dilutions	Assay Run: 08Aug14	Assay Run: 14Aug14	Assay Run: 26Aug14	Assay Run: 27Aug14
0	BLQ	BLQ	BLQ	BLQ
10	BLQ	BLQ	BLQ	BLQ
100	BLQ	BLQ	BLQ	BLQ
1,000	BLQ	BLQ	BLQ	BLQ

BLQ – Below Limit of Quantitation (3.9 ng/mL)

To demonstrate recovery of LT3114 from the citrate buffer formulation, the same dilution series was spiked with 5 and 10 ng/mL of LT3114 human reference standard. These concentrations are at or near the LLOQ and the results demonstrated that citrate buffer did not interfere with detection and quantification of LT3114 at any dilution as all spike recoveries were within the acceptance criteria of 25% of nominal ([Table 3](#)).

Table 3: Recovery of LT3114 from citrate buffer formulation.

Assay Run: 14Aug14						
Citrate Buffer Dilutions	Nominal Spike	Result (ng/mL)	% Recovery	Nominal Spike	Result (ng/mL)	% Recovery
0	5	5.50	110	10	9.08	91
10	5	4.20	84	10	7.70	77
100	5	4.14	83	10	7.94	79
1,000	5	4.02	80	10	8.35	84
DBS Spike	5	3.88	78	10	7.53	75
Assay Run: 26Aug14						
Citrate Buffer Dilutions	Nominal Spike	Result (ng/mL)	% Recovery	Nominal Spike	Result (ng/mL)	% Recovery
0	5	6.05	121	10	11.17	112
10	5	5.00	100	10	9.16	92
100	5	5.00	100	10	9.55	96
1,000	5	5.06	101	10	9.09	91
DBS Spike	5	4.87	97	10	8.60	86
Assay Run: 27Aug14						
Citrate Buffer Dilutions	Nominal Spike	Result (ng/mL)	% Recovery	Nominal Spike	Result (ng/mL)	% Recovery
0	5	5.69	114	10	10.38	104
10	5	4.51	90	10	8.28	83
100	5	4.55	91	10	8.37	84
1,000	5	4.37	87	10	8.23	82
DBS Spike	5	4.33	87	10	8.30	83

Calibration Curve

The protocol specified that data from ≥ 6 calibration curves be evaluated and that the accuracy of the back calculated LT3114 concentration had to be within 20% ($\pm 25\%$ at the LLOQ and ULOQ) of nominal to provide acceptable results. A total of 16 individual assay run calibration curves with associated accuracy and precision measurements are provided in [Table 4](#).

The standards 4000, 2000, 1000, and 1.95 ng/mL provided unacceptable back-calculated LT3114 concentrations for 15 of 16, 6 of 16, 5 of 16, and 9 of 16 calibration curves, respectively. This variability is reflected in the poor accuracy and/or precision measurements for these standards levels. All determination for standards ranging from 3.9 ng/mL to 500 ng/mL met acceptance criteria for accuracy and precision. These data indicate that the quantifiable range of the calibration curve is 3.9 to 500 ng/mL with the LLOQ = 3.9 ng/mL and the ULOQ = 500 ng/mL.

Table 4: Calibration curve performance.

Assay Run	LT3114 Concentration (ng/mL)											
	4000.00	2000.00	1000.00	500.00	250.00	125.00	62.50	31.25	15.60	7.80	3.90	1.95
08Aug14	5233.67	2231.77	900.79	452.67	269.07	124.06	63.80	32.63	14.91	6.96	3.79	2.46
08Aug14	182300.07	3821.41	1451.00	528.59	224.86	130.01	65.10	31.94	15.26	7.02	3.82	2.33
14Aug14	4962.52	2777.37	920.68	536.77	265.27	121.11	63.03	32.23	15.19	7.40	3.72	2.34
14Aug14	5534.16	1958.71	994.79	488.52	256.32	122.17	65.17	32.09	14.47	7.18	3.97	2.42
15Aug14	8660.97	2386.60	877.68	538.92	253.26	131.67	61.14	31.75	14.51	7.06	3.85	2.76
15Aug14	5898.44	1832.31	958.19	493.16	260.58	122.63	63.52	32.68	13.87	7.45	4.15	2.34
26Aug14	4863.79	2361.84	892.00	508.83	256.08	124.27	63.08	32.96	14.19	6.96	3.90	2.66
27Aug14	4921.66	2011.29	892.00	511.99	262.85	122.40	64.21	32.31	14.03	7.22	3.95	2.50
17Sep14	6631.73	2358.56	918.95	525.14	258.48	121.70	63.08	33.31	14.90	6.74	3.99	2.46
17Sep14	11673.10	2745.14	1281.59	500.82	257.95	118.54	61.47	34.01	15.69	7.35	3.67	1.99
25Sep14	6662.64	2280.25	896.66	459.36	280.35	122.29	61.61	33.26	14.31	7.17	3.92	2.53
25Sep14	5199.80	2135.40	983.08	467.07	263.50	127.35	61.77	32.37	15.06	7.17	3.83	2.38
26Sep14	5309.39	2698.68	1275.85	479.98	245.38	133.09	60.52	32.60	14.97	7.01	4.01	2.41
26Sep14	5350.53	1793.58	944.93	521.86	254.28	123.10	62.85	32.54	14.65	6.96	4.30	2.31
07Oct14	3588.84	3579.17	1248.73	516.83	263.65	118.64	62.51	33.64	14.94	7.45	3.78	2.17
13Oct14	2805.22	6009.1	2652.77	451.27	282.36	116.61	61.35	34.67	14.77	7.45	3.86	2.02
Mean LT3114 (ng/mL)	16849.78	2686.32	1130.61	498.86	259.64	123.73	62.76	32.81	14.73	7.16	3.91	2.38
Mean %RE	-321.2	-34.3	-13.1	0.2	-3.8	1.0	-0.4	-5.0	5.6	8.2	-0.2	-22.1
%CV	262.1	39.2	39.3	5.9	5.1	3.8	2.2	2.4	3.3	3	4.1	8.5

Gray highlight indicates that the back-calculation of the standard was not acceptable (>20% from nominal)

%RE = 100-(result/nominal) is a measure of accuracy and was calculated for each point of each curves and the mean results.

%CV = Standard deviation/Mean*100 is a measure of precision and was calculated for each standard concentration.

Accuracy and Precision: Intra- and Inter-assay

Accuracy (within 20% of nominal) and precision ($\%CV \leq 20\%$) of the LT3114 ELISA assay was based on the determination of three QC samples (500, 150 and 5 ng/mL) that spanned the range of the calibration curve. Intra-assay (6 analyses of each of 3 QC samples in ≥ 1 assay run; 1 assay run for QC1-500 ng/mL, 2 assays for QC2-150 ng/mL and QC3-5 ng/mL) and inter-assay (single analysis of each of 3 QC samples in 7 assay runs for QC1-500 ng/mL, 13 assay runs for QC2-150 ng/mL, and 13 assay runs for QC3-5 ng/mL) accuracy and precision were determined.

Intra-assay accuracy and precision: A total of 6 independent sets (duplicate determination in a single assay run) of QC1-500 ng/mL were assayed in a single assay run to assess intra-assay precision and accuracy. Intra-assay accuracy and precision were assessed using 6 independent sets of QC2-150 ng/mL and QC1-5 ng/mL in 2 independent assay runs. The intra-assay accuracy and precision of the 500, 150, and 5 ng/mL QCs are shown in [Table 5](#), [Table 6](#), and [Table 7](#), respectively. Both accuracy and precision were acceptable for all individual determinations of the three QCs as well as the intra-assay QC means.

Table 5: Intra-assay accuracy and precision of the 500 ng/mL QC.

Intra-assay - QC 1 (500 ng/mL)					
Assay Run	LT3114 (ng/mL)	Mean LT3114 (ng/mL)	SD	%CV	%RE
26Aug14	453.32	458.04	50.25	11.0	8.4
26Aug14	430.67				
26Aug14	415.61				
26Aug14	515.14				
26Aug14	409.13				
26Aug14	524.37				

%RE – percent relative error is a measure of accuracy [%RE = 100 – (result/nominal)].

Table 6: Intra-assay accuracy and precision of the 150 ng/mL QC.

Intra-assay - QC 2 (150 ng/mL)					
Assay Run	LT3114 (ng/mL)	Mean LT3114 (ng/mL)	SD	%CV	%RE
14Aug14	138.45	122.25	8.75	7.2	18.5
14Aug14	122.95				
14Aug14	122.31				
14Aug14	112.74				
14Aug14	119.32				
14Aug14	117.71				
15Aug14	133.26	124.40	6.38	5.1	17.1
15Aug14	126.82				
15Aug14	121.37				
15Aug14	115.16				
15Aug14	121.42				
15Aug14	128.39				

%RE – percent relative error is a measure of accuracy [%RE = 100 – (result/nominal)].

Table 7: Intra-assay accuracy and precision of the 5 ng/mL QC.

Intra-assay - QC 3 (5 ng/mL)					
Assay Run	LT3114 (ng/mL)	Mean LT3114 (ng/mL)	SD	%CV	%RE
14Aug14	4.60	4.20	0.22	5.3	16.0
14Aug14	4.19				
14Aug14	4.29				
14Aug14	4.04				
14Aug14	4.04				
14Aug14	4.04				
15Aug14	4.63	4.50	0.15	3.3	10.1
15Aug14	4.65				
15Aug14	4.33				
15Aug14	4.53				
15Aug14	4.55				
15Aug14	4.30				

%RE – percent relative error is a measure of accuracy [%RE = 100 – (result/nominal)].

Inter-assay accuracy and precision: Reproducibility was determined by calculating the between-run accuracy and precision of the three QCs from multiple independent assay runs. The data are presented in [Table 8](#). All three QCs showed acceptable accuracy and precision within the respective assays. Calculation of the mean LT3114 concentrations at each QC level across the multiple assays demonstrated acceptable inter-assay accuracy and precision.

Table 8: Inter-assay accuracy and precision of the LT3114 ELISA.

Inter-assay - QC1 (500ng/ml)				Inter-assay - QC2 (150ng/ml)				Inter-assay - QC3 (5ng/ml)			
Assay Run	LT3114 (ng/mL)	CV%	% Accuracy	Assay Run	LT3114 (ng/mL)	CV%	% Accuracy	Assay Run	LT3114 (ng/mL)	CV%	% Accuracy
				14Aug14	149.96	1.6	100	14Aug14	4.76	0.6	95
				15Aug14	165.77	1.4	111	15Aug14	5.46	5.2	109
				26Aug14	158.66	2.1	106	26Aug14	5.20	1.5	104
26Aug14	522.22	4.9	104	26Aug14	198.53 ^b	10.3	132	26Aug14	6.50 ^a	*35.1	130
27Aug14	429.58	2.3	86	27Aug14	158.93	3.6	106	27Aug14	5.81	21.3	116
17Sep14	507.19	0.7	101	17Sep14	143.36	3.1	96	17Sep14	4.41	0.0	88
				17Sep14	144.67	0.9	96	17Sep14	4.77	4.0	95
25Sep14	372.67	0.9	75	25Sep14	137.00	0.4	91	25Sep14	4.71	6.1	94
25Sep14	333.79 ^a	1.1	67	25Sep14	119.43	1.4	80	25Sep14	5.00	*33.6	100
26Sep14	439.17	1.2	88	26Sep14	147.38	1.9	98	26Sep14	4.80	1.4	96
26Sep14	401.34	0.9	80	26Sep14	173.19	0.3	115	26Sep14	5.29	3.7	106
				07Oct14	148.95	0.9	99	07Oct14	4.39	0.6	88
				13Oct14	147.18	0.2	98	13Oct14	4.68	1.7	94
Mean	429.42			Mean	153.31			Mean	5.06		
SD	68.20			SD	19.03			SD	0.60		
%CV	15.9			%CV	12.4			%CV	11.7		
%RE	14.1			%RE	-2.2			%RE	-1.2		

%RE – percent relative error is a measure of accuracy [%RE = 100 – (result/nominal)].

Black highlight indicates that QC was not analyzed in the assay.

*Precision of duplicates outside of acceptance range of $\leq 20\%$ CV.

Result outside of acceptance range ^a $\leq 25\%$ nominal (ULOQ and LLOQ) and ^b $\leq 20\%$ nominal.

Total error for intra-assay and inter-assay accuracy and precision is provided in [Table 9](#). The total error for all three QCs was acceptable for intra-assay and inter-assay accuracy and precision.

Table 9: Summary of intra-assay and inter-assay accuracy, precision, and total error.

Parameter	Statistic	QC Concentration (ng/mL)		
		500	150	5
Accuracy	Intra-assay %RE	8.4	17.8	13.1
	Inter-assay %RE	14.1	-2.2	-1.2
Precision	Intra-assay %CV	11.0	6.2	4.3
	Inter-assay %CV	15.9	12.4	11.7
Total Error	Intra-assay $ \%RE + \%CV$	19.4	24.0	17.4
	Inter-assay $ \%RE + \%CV$	30.0	14.6	12.9

Determination of the range of quantification of the assay was one goal for this validation. The three QCs were selected in the protocol prior to performance of the validation and establishing the range of quantification. Initial assay runs demonstrated that the QC-750 ng/mL provided variable results and the protocol was amended to lower the high QC to 500 ng/mL. Results of this validation show that the ULOQ for the ELISA method is 500 ng/mL. Therefore, the high QC for studies that utilize this validated method will be lowered to 300 ng/mL and this change is reflected in the ELISA method acceptance criteria provided in [Conclusions](#) section.

Linearity of Dilution

Dilution linearity was demonstrated by analyzing a 2-fold dilution series of the LT3114 in a citrate buffer formulation (Lot# 103984) from 1:40,000 to 1:1,280,000 in two independent assay runs. Accuracy (within 20% of nominal) and precision ($\%CV \leq 20\%$) were determined for each dilution and then the linearity of dilution was calculated based on the value from the previous dilution divided by 2 and on the accuracy of the result adjusted for dilution factors. The results are provided in [Table 10](#). Results demonstrated acceptable linearity of dilution when compared to the previous dilution result and also to nominal for all dilutions except at the 1:40,000. The OD₄₅₀ absorbance values for the 1:40,000 dilution were outside of the quantifiable range (3.9 – 500 ng/mL) of the calibration curve, as determined by this validation. Therefore, the results for 1:40,000

dilution of the LT3114 in a citrate buffer formulation are not accurate indicating the requirement for further dilution of the neat LT3114 in a citrate buffer formulation. The accuracy, precision, and total error for the mean results for dilutions from 1:80,000 to 1:640,000 were acceptable. Accuracy and precision were acceptable for the 1:1,280,000 dilution, however, total error was 31.4%, which was slightly greater than the acceptance criteria of 30%.

Table 10: Linearity of dilution for LT3114 in a citrate buffer formulation.

Nominal (mg/ml)	Dilution	14Aug14 Assay Run Results		17Sep14 Assay Run Results		Average		Recovery (%)		Precision	Total Error
		Result	Adjusted Result (mg/mL)	Result	Adjusted Result (mg/mL)	Mean Result	Mean Adjusted Result (mg/mL)	From Previous Result	Relative to nominal	%CV	$ \%RE ^1 + \%CV$
32.1	40000	786.88	31.48	1117.40	44.70 ²	952.14	38.09	N/A	119	24.5 ³	43.5 ⁴
	80000	350.25	28.02	408.30	32.66	379.28	30.34	80	95	10.8	15.8
	160000	164.94	26.39	183.28	29.33	174.11	27.86	92	87	7.4	20.4
	320000	80.21	25.67	92.86	29.71	86.53	27.69	99	86	10.3	24.3
	640000	38.79	24.82	47.73	30.55	43.26	27.69	100	86	14.6	28.6
	1280000	18.80	24.06	23.73	30.37	21.26	27.21	98	85	16.4	31.4 ⁴

¹%RE = % Relative Error calculated as $[\%RE = 100 - (\text{result}/\text{nominal})]$,

Outside of acceptance criteria for:

²accuracy of within 20% of nominal,

³precision of $\leq 20\%$, and

⁴total error of $\leq 30\%$.

Stability

The impact of multiple freeze/thaw cycles (up to 4 cycles), holding the samples at room temperature for ≥ 4 hours, storing the QC1-500 ng/mL and QC3-5 ng/mL at $\leq -70^{\circ}\text{C}$ for ≤ 3 months, and storing LT3114 in a citrate buffer formulation at neat (32.1 mg/mL) and 2 mg/mL at $2-8^{\circ}\text{C}$ for ≤ 3 months. Acceptable accuracy (within 20% of nominal) was used to confirm assay stability of LT3114 under these conditions.

These parameters were examined using aliquots of the QC1, QC3, and/or LT3114 in a citrate buffer formulation tested against a calibration curve of freshly prepared standards. The accuracy and precision of the results were calculated to determine the stability of the LT3114 under these conditions. The stability results of LT3114 stored frozen are provided in [Table 11](#). The stability results for LT3114 stored at $2-8^{\circ}\text{C}$ are provided in [Table 12](#).

Table 11: Stability of LT3114 when stored at RT, $\leq -70^{\circ}\text{C}$, and following freeze-thaw.

Assay Run	QC1 (500 ng/mL) ¹			QC3 (5 ng/mL) ²		
	Condition	Result (ng/ml)	% Accuracy	Condition	Result (ng/ml)	% Accuracy
07Oct14	1X	513.05	103	1X	4.54	91
07Oct14	2X	385.16	77*	2X	4.39	88
07Oct14	3X	396.52	79*	3X	4.37	87
07Oct14	4X	509.75	102	4X	4.41	88
26Aug14	Ambient	585.51	117	Ambient	5.04	101
17Sep14	Frozen Stability (26 days)	449.46	90	Frozen Stability (35 days)	4.47	89

*500 ng/mL is the ULOQ, therefore, acceptable accuracy is within 25% of nominal.

QCs prepared on ¹22 Aug 14 and ²13 Aug 14.

Table 12: Storage stability of LT3114 in a citrate buffer formulation at $2-8^{\circ}\text{C}$.

Assay Run Date*	Sample	Nominal (ng/mL)	Result (mg/mL)	% Accuracy
14Aug14	Neat	32.1	26.74	83
17Sep14	Neat	32.1	32.89	102
07Oct14	Neat	32.1	27.01	84
13Oct14	2.0 mg/mL ¹	2.0	1.71	86

*Result for 14-Aug-14 and 17-Sep-14 are the average of the linearity of dilution results

¹2.0 mg/mL stock prepared on 13Aug14 and stored at $2-8^{\circ}\text{C}$

The impact of freeze-thaw cycles on the stability of LT3114 was assessed by thawing and refreezing QC1-500 ng/mL and QC3-5 ng/mL up to 4 times with at least 12 hours between each cycle. At the time the study protocol was finalized, the calibration curve range was defined as 1.95 – 4000 ng/mL, as the method transferred from the Sponsor (BRT 20140420). Therefore, it was assumed that the QCs would be below the ULOQ and above the LLOQ. The results of the calibration curve assessment (see the [Calibration Curve](#) section) indicated that the quantifiable range of the assay is 3.9 to 500 ng/mL, resulting in QC1-500 ng/mL being at the ULOQ and QC3-5 ng/mL at the LLOQ. Therefore, acceptable accuracy for these QC samples is within 25% of nominal. Concentrations for the QCs were acceptable for all four freeze-thaw cycles.

Frozen storage stability was also assessed using QC1-500 ng/mL and QC3-5 ng/mL. These QCs were stored at $\leq -70^{\circ}\text{C}$ for 26 days (QC1-500 ng/mL) and 35 days (QC3-5 ng/mL) prior to determination of LT3114 concentration. Frozen storage was acceptable for both QCs ([Table 11](#)). Additional analyses were performed using these QCs resulting in total frozen storage stability of 35 days and 61 days at $\leq -70^{\circ}\text{C}$ for the QC1-500 ng/mL and QC3-5 ng/mL, respectively.

The stability of LT3114 at ambient temperature was assessed by storing QC1-500 ng/mL and QC3-5 ng/mL on the benchtop for 4 hours and 6 minutes prior to being prepared and analyzed for LT3114 concentration. The concentrations obtained for both QCs showed acceptable accuracy, within 20% of nominal.

The stability of LT3114 in a citrate buffer formulation was determined following storage at 2-8°C for 46, 80, and 99 days from manufacture for the bulk material and for 60 days following formulation of a 2 mg/mL stock. These durations are based on the assumption that the test article was produced on the last day of June 2014 as the day of the month is not specified in the certificate of analysis ([Appendix I](#)). LT3114 in a citrate buffer formulation demonstrated acceptable storage stability under these conditions as indicated by accuracy within 20% of nominal concentrations [Table 12](#).

Robustness/Ruggedness

Robustness of the ELISA method was determined using two parameters; 1) reading an assay run on different plate readers (Spectramax 340 and M2e), 2) performing analyses

by two independent operators. These parameters were considered appropriate if the accuracy (within 20% of nominal) and precision (%CV $\leq$ 20%) of the standard curve and QCs were acceptable.

The impact of using two microplate spectrophotometers to quantify the same plate is shown in Table 13 for the calibration curve and QCs. The calibration curve showed acceptable accuracy and precision for the quantifiable range of 3.9 – 500 ng/mL for both plate readers. The QC2-150 ng/mL and the QC3-5 ng/mL QCs showed acceptable accuracy and precision on both plate readers. The QC3-500 ng/mL was not analyzed using two independent plate readers. This demonstrates that results generated by both plate readers are acceptable.

Table 13: Impact of two independent plate readers on results of a single assay.

	Spectramax 340	Spectramax M2e			
Standard (ng/mL)	LT3114 (ng/mL)	LT3114 (ng/mL)	Mean	%CV	% of nominal
Operator 1; Assay Run: 15Aug14					
4000.00	5437.40	5898.44	5667.92	5.8	142
2000.00	1940.49	1832.31	1886.40	4.1	94
1000.00	984.30	958.19	971.25	1.9	97
500.00	481.01	493.16	487.09	1.8	97
250.00	261.32	260.58	260.95	0.2	104
125.00	122.29	122.63	122.46	0.2	98
62.50	64.12	63.52	63.82	0.7	102
31.25	32.68	32.68	32.68	0.0	105
15.60	13.77	13.87	13.82	0.5	89
7.80	7.36	7.45	7.41	0.8	95
3.90	4.05	4.15	4.10	1.7	105
1.95	2.46	2.34	2.40	3.3	123
QC2-150	124.40	125.36	124.88	0.5	83
QC3-5	4.50	4.46	4.48	0.6	90
Operator 2; Assay Run Date: 15Aug14					
Standard (ng/mL)	LT3114 (ng/mL)	LT3114 (ng/mL)	Mean	%CV	% of nominal
4000.00	11253.87	8660.97	9957.42	18.4	249
2000.00	2099.24	2386.60	2242.92	9.1	112
1000.00	892.11	877.68	884.90	1.2	89
500.00	552.27	538.92	545.60	1.7	109
250.00	251.20	253.26	252.23	0.6	101
125.00	132.18	131.67	131.93	0.3	106
62.50	60.84	61.14	60.99	0.3	98
31.25	31.77	31.75	31.76	0.0	102
15.60	14.61	14.51	14.56	0.5	93
7.80	7.01	7.06	7.04	0.6	90
3.90	3.87	3.85	3.86	0.3	99
1.95	2.76	2.76	2.76	0.0	142
QC2-150	165.77	164.06	164.91	0.7	110
QC3-5	5.46	5.58	5.52	1.6	110

CONCLUSIONS

The present study was conducted to validate an ELISA method for the quantification of LT3114 in a citrate buffer formulation. Validation parameters that were examined included determination of LOD, citrate buffer interference, calibration curve performance, accuracy, precision, linearity, stability evaluation, and robustness/ruggedness. The results demonstrate the validation goals were achieved and the assay is suitable for the purpose of detecting LT3114 in a citrate buffer formulation. Acceptance criteria for analysis of dose formulation samples will be the following:

Calibration Curve Acceptance Criteria:

- Calibration curve range: 3.9 to 500 ng/mL
 - Accuracy of standards: 20% of nominal (25% at LLOQ and ULOQ)
 - Precision of the standards will be within 20%
 - At least 75% of the standards (minimum of 6 points) will meet these criteria.
- Failed standards will be removed from the analysis.

QC Sample Acceptance Criteria:

- QC1-300 ng/mL; QC2-150 ng/mL; QC3-5 ng/mL
- Accuracy of QC samples: 20% of nominal (25% at 5 ng/mL and 300 ng/mL)
- Precision of the QCs will be within 20% (25% at 5 ng/mL and 300 ng/mL)
- At least 67% of all QC samples will meet these criteria. In addition, at least 50% of the QC samples at each level will meet these criteria.
- Total error ($|\%RE| + \%CV$) of each QC will be $\leq 30\%$ ($\leq 40\%$ at 5 ng/mL and 300 ng/mL)

Study Dose Formulation Sample Acceptance Criteria:

- Precision of the replicates will be within 20% (25% at LLOQ and ULOQ)
- Sample concentrations will be acceptable when accuracy is within 20% of nominal

Study Dose Formulation Sample Reanalysis:

- Samples will be reanalyzed if the assay is rejected due to failure of the standards or QCs to meet acceptance criteria provided above.

- Samples will be reanalyzed if the mean OD₄₅₀ for the replicates is outside the range of OD₄₅₀ values for the calibration curve. The sample will be diluted appropriately and reanalyzed. This does not apply if samples at the MRD are below the range of the standard curve.
- For samples that are within the range of the standard curve, samples will be reanalyzed if the precision of the replicates is not acceptable.

DOCUMENTATION AND RETENTION OF DATA

Description of Circumstances Affecting Data Quality or Integrity

BRT is not aware of any circumstances surrounding this study that would adversely affect data quality or integrity.

Deviations

The following deviations occurred during the performance of this study.

- 1) The original high QC was set at 750 ng/mL. During performance of the validation, it was determined that this was outside the quantifiable ranges of the assay. The Study Director adjusted the high QC to 500 ng/mL LT3114. However, the analyst prepared the QC1-500 ng/mL prior to finalization of the protocol amendment. There is no impact of this deviation as the QC was prepared as requested by the Study Director and was not analyzed until after the amendment was executed.
- 2) The analyst prepared the free-thaw samples according to the protocol but without adequate documentation. There is no impact of this deviation since a new set of freeze-thaw samples were prepared with corresponding documentation and used to assess stability.
- 3) A technical error occurred during the assay run performed on 08Aug14. A 50-fold intermediate dilution step was missed during preparation of the human reference LT3114 prior to spiking into citrate formulation buffer. The data from the affected assay was not reported and the spike-recovery experiment was repeated. Therefore, there was no impact of this deviation on the study.
- 4) Frozen stability of LT3114 was assessed at 35 days of storage rather than at ≤ 4 weeks as stated in the study protocol due to technical oversight. Since this storage duration is longer than specified in the protocol and the results were acceptable, there is no impact of this deviation on the study.
- 5) The secondary antibody was diluted in blocking solution for the first dilution step as per protocol. However, the second dilution step was also performed in blocking buffer rather than in DBS as specified in the protocol. This was done throughout the performance of the validation. Since the parameters examined met acceptance criteria and the validation demonstrated that the ELISA method is suitable for quantifying LT3114 in citrate formulation buffer, there is no impact of this deviation. The ELISA method provided in [Appendix VIII](#) that will be used for dose formulation analysis reflects the change in procedure for preparing the secondary antibody.
- 6) The zero standard was not used for the calibration curve throughout the validation but was listed in the ELISA method in the protocol. This was a typographical error and the intention of the Study Director was to use the zero as a blank. There

is no impact of this deviation on the study and the quantification range of the ELISA methods as determined in this validation is provided in the method.

Maintenance of Raw Data and Records

The experimental data and the final report will be retained in the archives of the Testing Facility for a period of at least 2 years from the date that the final report is issued. After that time, or as per contract agreement, the Sponsor will be notified and the data and report will be returned to the Sponsor, unless alternative arrangements are made. BRT assay and equipment logbooks will be retained and stored in the archives of the Testing Facility.

APPENDICIES

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Appendix I: Certificate of Analysis for Lpathomab-DS, Lot# Gallus-MO 103948.

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
		PAGE	1 of 2

Name	Part Number		Lot Number
Lpathomab-DS, 30 mg/mL	TBD		GA121459
Formulation Buffer			Storage Temperature
24.3 mM Sodium Citrate, 51.4 mM Sodium Phosphate, 300 mM D-Trehalose, 200 ppm Polysorbate-80, pH 5.0			2 to 8°C
Production Date	Test Date	Retest Date	Intended Use
JUN 2014	JUL 2014	DEC 2014	GLP-conforming Toxicity Testing
Manufacturer	Manufacturer Part Number		Manufacturer Lot Number
Gallus-MO, St. Louis, MO	TBD		Gallus-MO 103948
Attribute	Test	Specification	Results
Identity	Appearance	Particulate free, clear, colorless to pale yellow, slightly opalescent solution	Particulate free, clear, slightly yellow solution
	pH	5.0 ± 0.2	5.1
	CGE (Non-reduced)	Major band at 150 kDa	Conforms
	CGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~ 25 kDa	Conforms
	iCE	Comparable to RS	Conforms
Minor Peak at pI ~8.3 Major Peak at pI ~8.4 Minor Peak at pI ~8.6 Faint Peak at pI ~8.7		15% 60% 13% 4%	
Potency	API Concentration by OD at 280 nm	30 ± 3 mg/mL	32.1 mg/mL
	LPA Binding	100% ±40% of RS	89%

Confidential

Lpath Inc.		Certificate of Analysis	
TITLE	Lpathomab-DS, 30 mg/mL, for use in GLP-conforming testing		REVISION No.
			1.0.0
			PAGE 2 of 2

Attribute	Test	Specifications		Results
Purity	CGE non-reduced	Monomer + Non-glycosylated Monomer	≥ 90%	98.8%
		Non-glycosylated Monomer	Report %	2.6%
		Other impurities	No single impurity > 5%	conforms
	CGE, reduced	HC + LC + Non-Glycosylated HC	≥ 90%	99.0%
		HC	Report %	66.6%
		Non-Glycosylated HC	Report %	1.0%
		LC	Report %	31.4%
		Other Impurities	No single impurity > 5%	0.4% (high molecular weight)
	SE-HPLC	Purity [%]	≥ 90%	99.5%
		Aggregates [%]	≤ 5%	0.5%
		Other Impurities [%]	No single impurity > 5%	None detected

Attribute	Test	Specification	Results
Safety	Residual Protein-A	≤ 10 ppm	<1 ppm
	Residual DNA	≤ 5 pg/mg	<0.05 pg/mg
	CHO HCP	≤ 100 ppm	15 ppm
	Endotoxin, USP <85>	Report	<0.06 EU/mL ¹
	Bioburden; USP <61> (Membrane Filtration)	≤ 5 CFU/10 mL	None detected

JUDGMENT

Based on a comprehensive review of the lot specific manufacturing and testing records, this API is approved for use in GLP-conforming pre-clinical animal toxicity tests.

Caution! Not for use in humans!

Quality Assurance:



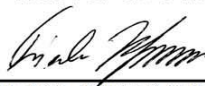
Acting Head of QA - Frieder K. Hofmann, PhD

Date: 22AUG2014

¹ At 0.06 EU/mL and a tolerance of 5 EU/kg animal weight, the maximum volume that may be dosed is 83 mL/kg or 2,675 mg Lpathomab per kg of animal.

Confidential

Appendix II: Certificate of Analysis for the Citrate Formulation Buffer (Lot# 104011-001).

Lpath Inc.		Certificate of Compliance			
TITLE	24.3 mM Citrate, 51.4 mM Sodium Phosphate Dibasic, 300 mM D-Trehalose, pH 5.0 Buffer				
					PAGE 1 of 1
Manufacturer			Lot Number		
Gallus Biopharmaceuticals, St. Louis, MO			104011-001		
Production Date		Expiry Dates¹			Intended Use
JUN 2014		Storage: 15 to 30°C; SEP 2014	Storage: 2 to 8°C; MAR 2015	Storage: -25 to -15°C; JUN 2016	Storage: -90°C to -60°C; JUN 2019
					GLP-conforming toxicity testing and Lpathomab sample diluent
Attribute	Specification				Results
Appearance	Particulate free, clear, colorless				Conforms
pH	4.9 to 5.1				5.1
Sterility	Confirmed integrity of sterilizing membrane filter				Conforms
JUDGMENT					
Based on a comprehensive review of the executed production batch record, this buffer is approved for use in GLP-conforming toxicity testing and as sample diluent for Lpathomab. Caution – Not for use in humans Quality Assurance:  Date: 18JUL2014 Acting Head of QA - Frieder K. Hofmann, PhD					

¹ Buffer not stored < -60°C or used for toxicity testing, requires aseptic handling.

Confidential

Appendix III: Certificate of Analysis for the Human Reference LT3114 (HU-anti-LPA antibody Lot# LP121416).

		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	1 of 5

Name	Part Number	Lot Number
Humanized Lpathomab	LT3114	LP121416

Formulation Buffer	Storage Temperature
PBS (Mediatech #21-040, Lot# 21040283)	2 to 8 °C

Production Date	Retest Date	Project /Use
5-10-2013	N/A	TBD

Summary
<u>Production location, date, and method</u> The antibody was produced from stable CHO clone (191+192) 3114 3B8/1B4/2F1 in 4x 3-L shake flasks at Lpath. Takedown on d10, May 10, 2013. Pooled, clarified harvest was stored at -20 °C. Lot LP121416 consists of 3x 1-L bottles of supe which cracked during storage. Lot LP121417 consists of 1x 1-L bottle of supe which did not crack during storage. Purified lots LP121416 and LP121417 were pooled to create lot LP121419. <u>Purification location, date and method</u> Antibody purification was initiated on August 21, 2013 at Lpath using affinity and AEX chromatography (MabSelect Protein A and Q membrane Sartorius #92IEXQ42D4-OOA, respectively). Following concentration, the antibody was dialyzed against 3x 2-L PBS (Mediatech) and passed through a 0.22 um filter flask unit (Corning). <u>Concentration by A₂₈₀ and extinction coefficient</u> The final sample is 21.5 mg/mL as determined by UV spectroscopy using a theoretical extinction coefficient of 1.4 mL/mg. <u>Total amount of Ab (mg)</u> 1400 mg <u>Appearance details:</u> Clear, slightly yellow solution

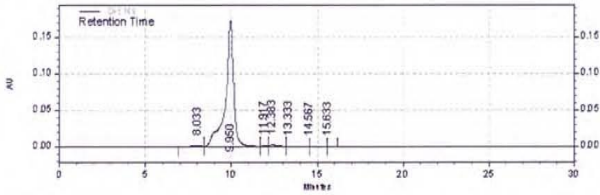
Confidential

Lpath INCORPORATED		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	2 of 5

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear solution Colorless or slightly yellow	Particulate free, clear solution Slightly yellow
	SDS-PAGE (Non-Reduced)	Report Results	Major band at 165 kDa
	SDS-PAGE (Reduced)	Two pronounced bands; One at ~50 kDa; One at ~25 kDa	Two pronounced bands; One at ~55 kDa; One at ~28 kDa
	Insert images of SDS-PAGE	Non-Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty  Reduced LT3114 new 2.7 µg LT3114 new 2.7 µg Standard Empty 	
		Isoelectric Focusing (by IEF Gel)	Report Results

Confidential

		Certificate of Analysis Form			
TITLE Certificate of Analysis for Monoclonal Antibody Produced by Lpath					LOT NUMBER LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	3 of 5

Attribute	Test	Specifications	Results																																													
Potency	Protein Concentration (OD-280)	Report Results	20.14 mg/mL																																													
	LPA Binding Direct ELISA (EC 50)	Report Results	25.05 ng/mL																																													
	Bioactivity Assay (IL-6 and EDG2)	Positive Result	Does not stimulate IL-6 EDG2 IC50 = 13.65 µg/mL																																													
Purity	Insert image of SE-HPLC  <table border="1"> <caption>Det 166 Results</caption> <thead> <tr> <th>Time</th> <th>Area</th> <th>Area %</th> <th>Height</th> <th>Height %</th> </tr> </thead> <tbody> <tr><td>8.033</td><td>68097</td><td>1.14</td><td>1413</td><td>0.78</td></tr> <tr><td>9.950</td><td>5707101</td><td>95.95</td><td>173377</td><td>96.24</td></tr> <tr><td>11.917</td><td>35431</td><td>0.60</td><td>1744</td><td>0.97</td></tr> <tr><td>12.383</td><td>97022</td><td>1.63</td><td>2645</td><td>1.47</td></tr> <tr><td>13.333</td><td>31081</td><td>0.52</td><td>668</td><td>0.37</td></tr> <tr><td>14.567</td><td>8116</td><td>0.14</td><td>234</td><td>0.13</td></tr> <tr><td>15.633</td><td>1015</td><td>0.02</td><td>62</td><td>0.03</td></tr> <tr> <td>Totals</td> <td>5947863</td> <td>100.00</td> <td>180143</td> <td>100.00</td> </tr> </tbody> </table>			Time	Area	Area %	Height	Height %	8.033	68097	1.14	1413	0.78	9.950	5707101	95.95	173377	96.24	11.917	35431	0.60	1744	0.97	12.383	97022	1.63	2645	1.47	13.333	31081	0.52	668	0.37	14.567	8116	0.14	234	0.13	15.633	1015	0.02	62	0.03	Totals	5947863	100.00	180143	100.00
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Totals	5947863	100.00	180143	100.00																																												
SE-HPLC	<table border="1"> <tr> <td>Purity [% as IgG monomer]</td> <td>≥ 90%</td> <td>96%</td> </tr> <tr> <td>Aggregates [as % of total peak area]</td> <td>≤ 6%</td> <td>1%</td> </tr> <tr> <td>Other Impurities [%]</td> <td>No individual impurity ≥ 3%</td> <td>No individual impurity ≥ 3%</td> </tr> </table>	Purity [% as IgG monomer]	≥ 90%	96%	Aggregates [as % of total peak area]	≤ 6%	1%	Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%																																						
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Confidential

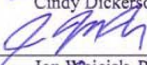
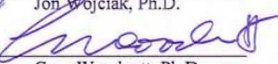
		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	4 of 5

Attribute	Test		Specifications	Results
	SDS-PAGE reduced	Purity [%]	≥ 90%	≥ 90%
		50 kDa Band	~65%	~67%
		25 kDa Band	~35%	~33%
		Other Impurities [%]	No individual impurity ≥ 3%	No individual impurity ≥ 3%
	Endotoxin		≤ 10.0 EU/mg of protein	0.06 EU/mg
	Bioburden		≤ 10 CFU/mL	Not done

Comments																																
Source Data: <table border="1"> <thead> <tr> <th></th> <th>Lpath notebook</th> <th>Pages</th> </tr> </thead> <tbody> <tr> <td>Production</td> <td>277</td> <td>149</td> </tr> <tr> <td>Purification</td> <td>296</td> <td>11,18,25</td> </tr> <tr> <td>UV A280</td> <td>296</td> <td>48</td> </tr> <tr> <td>SDS-PAGE</td> <td>296</td> <td>26</td> </tr> <tr> <td>LPA binding ELISA</td> <td>221</td> <td>21-22</td> </tr> <tr> <td>IL-6</td> <td>237</td> <td>48-49</td> </tr> <tr> <td>EDG2</td> <td>237</td> <td>51</td> </tr> <tr> <td>SE-HPLC</td> <td>296</td> <td>33-34</td> </tr> <tr> <td>Endotoxin</td> <td>298</td> <td>13</td> </tr> </tbody> </table> Additional Comments on Sample Quality:				Lpath notebook	Pages	Production	277	149	Purification	296	11,18,25	UV A280	296	48	SDS-PAGE	296	26	LPA binding ELISA	221	21-22	IL-6	237	48-49	EDG2	237	51	SE-HPLC	296	33-34	Endotoxin	298	13
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Confidential

Lpath INCORPORATED		Certificate of Analysis Form			
TITLE	Certificate of Analysis for Monoclonal Antibody Produced by Lpath				LOT NUMBER
					LP121416
DOCUMENT NUMBER	REVISION	SUPERCEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-13	0.0.1	0.0.0	OCT 31, 2013	Nov 1, 2013	5 of 5

JUDGMENT
Approved for testing in humans ☐ - Accepted ☐ - Rejected For Research Only Not For Use in Humans Prepared by: <u></u> Date: <u>12-6</u> 2013 Cindy Dickerson, Ph.D. Reviewed by: <u></u> Date: <u>12/6/</u> 2013 Jon Wojciak, Ph.D. Approved by: <u></u> Date: <u>3/1/2014</u> 2013 Gary Woodnutt, Ph.D.

Confidential

Appendix IV: Memo from Sponsor for Human Reference LT3114 (HU-anti-LPA antibody Lot# LP121416) indicating stability parameters.



May 20, 2014

COPY
010ct14

Victor J. Johnson, Ph.D.
Burleson Research Technologies, Inc.
120 1st. Flight Ln
Morrisville NC 27560
Tel. 919-719-2500
Email: vjohnson@brt-labs.com

Dear Victor,

Enclosed, please find the following materials in this shipment:

40.2mg of LT3114 (HU-anti-LPA antibody, Lot#LP121416) aliquoted in 2 vials
1ml/each at 20.1 mg/ml

Upon arrival, please keep frozen. Dilutions should be made immediately, prior to using the material and should be stored at 4° C. Material stored at 4° C is stable for up to 6 months.

Should you have any questions regarding the material, please do not hesitate to contact us.

Best Regards,

A handwritten signature in black ink, appearing to read "Rosalia Matteo".

Rosalia Matteo, PhD
Asst. Director
Lpath Incorporated

4025 Sorrento Valley Blvd., San Diego, CA 92121-1404 • Main: (858) 678-0800 • Fax: (858) 678-0900
www.lpath.com

Appendix V: Certificate of Analysis for Coating Material details (12OLPA-SMCC-BSA Lot# LP121430).

Lpath Inc.			Certificate of Analysis		
TITLE	LPA-SMCC-BSA Coating Material				LOT NUMBER
					LP121430
DOCUMENT NUMBER	REVISION	SUPERSEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-11	0.0.0	0.0.0	AUG 28, 2008	AUG 28, 2008	1 of 2

Name	Part Number	Lot Number
12:0 LPA coating material	LPA-SMCC-BSA	LP121430
Formulation Buffer		Storage Temperature
PBS, (Cellgro Cat No. 21-040)		-70°C
Production Date	Retest Date	Project /Use
November 21 st , 2013	November 21 st , 2015	LPA-binding ELISA

Comments
<u>Description:</u> 12:0LPA-SH (Lpath, Cat No. C12 LPA-SH; Lot 1B) was conjugated to maleimide-activated BSA (Thermo Scientific, Cat No. 77115; Lot NG175286). The protein-LPA conjugate was purified using a protein desalting column (Thermo Scientific, Cat No. 89891; Lot OE186028). Concentration (of BSA) determined using the BCA assay (Thermo Scientific, Cat No. 23225; Lot MK166704): 5 mg/ml Total amount of material (mg): 15 mg

Attribute	Test	Specifications	Results
Identity	Appearance	Particulate free, clear colorless solution	Particulate free, clear colorless solution
	pH	6.8-8.0	7.4
Potency	BSA Concentration (BCA)	Report Results	5 mg/ml

Confidential

Lpath Inc.		Certificate of Analysis			
TITLE	LPA-SMCC-BSA Coating Material				LOT NUMBER
					LP121430
DOCUMENT NUMBER	REVISION	SUPERSEDED REVISION	ISSUE DATE	EFFECTIVE DATE	PAGE
FM-COA-11	0.0.0	0.0.0	AUG 28, 2008	AUG 28, 2008	2 of 2

Attribute	Test	Specifications	Results
	Antibody Detection Range EC ₅₀ -LPA-Binding ELISA with B3 Lot HR111266	Report Results	125 ng/ml

JUDGMENT	
☒ Accepted ☐ - Rejected Approved for testing in humans For Research Only Not For Use in Humans Prepared and Reviewed by (Quality Control): <u>Jon Wojniak, Ph.D.</u> Date: <u>4/15/2014</u> Approved by: <u>Gary Woodnutt, Ph.D.</u> Date: <u>4/15/2014</u>	

Confidential

Appendix VI: Memo from Sponsor for Coating Material details (12OLPA-SMCC-BSA Lot# LP121430).BRT20140420
BRT20140629A

June 4, 2014

Victor J. Johnson, Ph.D.
Burleson Research Technologies, Inc.
120 1st Flight Ln
Morrisville NC 27560
Tel. 919-719-2500
Email. vjohnson@bri-labs.com

COPY
01oct14

Dear Victor,

Enclosed, please find the following materials in this shipment:

1 mg of 12OLPA-SMCC-BSA (Coating material, Lot#LP121430) aliquoted in 2 vials
0.2ml/each at 5 mg/ml. Upon arrival, please Freeze for long term storage.
Material stored at 4° C is stable for up to 6 months.

~~N/A not used on day 10oct14~~
96.3mg of LT3114 (HU-anti-LPA antibody, Lot# P9023) aliquoted in 3 vials
1ml/each at 32.1 mg/ml. Upon arrival, please stored at 4° C. Material stored at
4°C under sterile condition is stable for up to 1 year.

Should you have any questions regarding the material, please do not hesitate to contact us.

Best regards,



Rosalia Matteo, PhD
Asst. Director
Lpath Incorporated

Appendix VII: Summary of all assay runs performed during validation of the method.

Assay Run #	Assay Run Date	Included in validation (Yes / No)	Comments
1	5-Aug-14	No	Run excluded due to insufficient mixing of secondary antibody into the glycerol diluent resulting in variability
2	7-Aug-14	No	Run excluded due to insufficient mixing of secondary antibody into the glycerol diluent resulting in variability
3	8-Aug-14	Yes	- Calibration curve data reported. - Citrate buffer matrix interference data reported. - Citrate buffer spike results were excluded due to a dilution error in preparation (see Deviation 3). - QC data not reported due to formulation in the incorrect lot of citrate formulation buffer.
4	14-Aug-14	Yes	- Calibration curve data reported. - QC2-150 ng/mL and QC3-5 ng/mL data reported. - QC1-750 ng/mL data not reported as QC1 was amended in the protocol to 500 ng/mL.
5	14-Aug-14	Yes	All data reported.
6	15-Aug-14	Yes	All data reported.
7	15-Aug-14	Yes	- Calibration curve data reported. - QC2-150 ng/mL and QC3-5 ng/mL data reported. - QC1-750 ng/mL data not reported as QC1 was amended in the protocol to 500 ng/mL.
8	26-Aug-14	Yes	All data reported.
9	26-Aug-14	Yes	- QC data were out of acceptance. QC data were reported as part of accuracy and precision assessment and were identified as out of acceptance range in the report. - Due to QCs being out of acceptance, other parameters investigated (DBS spike, citrate buffer spike, citrate buffer matrix background) were repeated and data not reported.
10	27-Aug-14	Yes	All data report.
11	17-Sep-14	Yes	- Calibration curve data reported. - QC1-500 ng/mL, QC2-150 ng/mL, and QC3-5 ng/mL freeze-thaw and stability data reported. - The original QC1-750 ng/mL was inadvertently run, data not reported as QC1 was amended in the protocol to 500 ng/mL.
12	17-Sep-14	Yes	- Calibration curve, QC, and linearity of dilution data reported. - Drug stability data not report due to suspected technical error. Subsequent assessments of stability were acceptable.
13	25-Sep-14	Yes	All data report.
14	25-Sep-14	Yes	All data report.
15	26-Sep-14	Yes	All data report.
16	26-Sep-14	Yes	All data report.
17	7-Oct-14	Yes	All data report.
18	13-Oct-14	Yes	All data report.

Appendix VIII: Method used for validation of the ELISA for detection of LT3114 in citrate formulation buffer.

METHOD

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Dilipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material should be kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material should be stored at 2-8° C for up to 4 months.
2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA l, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 500, 150, and 5 ng/mL. Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 1 $\mu\text{g/ml}$ in 0.05 M Carbonate Buffer.
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 $\mu\text{L/well}$ coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.
 - 3.2 Dispense 150 $\mu\text{L/well}$ of blocking solution to each well and cover the plate(s) with plate sealers.

BRT-BURLESON RESEARCH TECHNOLOGIES, INC.

- 3.3 Incubate plate(s) at 37°C for 1 hour $\pm$ 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300 μ l/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 μ L/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 μ l of appropriate sample to designated wells in duplicate. Add 100 μ l of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour $\pm$ 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300 μ l/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution (see [Deviation Section](#)).
 - 6.3 Dispense 100 μ l of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour $\pm$ 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 μ l **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.
8. Stop Solution Addition
 - 8.1 Dispense 100 μ l 1.0 M H₂SO₄ solution to each well.
 - 8.2 Measure absorbance at 450 nm on microplate reader.

BRT-BURLESON RESEARCH TECHNOLOGIES, INC.

Report Amendment

Sponsor: Lpath Incorporated

BRT Study No.: BRT 20140629A

Study Title: Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation

Report Amendment No.: 1

Original Report Date: 21 Nov 14

Report Amendment Date: 10 Dec 14

Detailed Description of Amendment

Page 32: DEVIATIONS Section should read;

“● The LPA-Conjugated-BSA was diluted to 2.5 µg/mL rather than 1 µg/mL as specified in the protocol. This was due to an error in the dilution schema on the reagent preparation form. All validation assay runs used the 2.5 µg/mL coating concentration and therefore, the ELISA method was successfully validated as such. There is no impact of this deviation on the study.”

Reason for Amendment

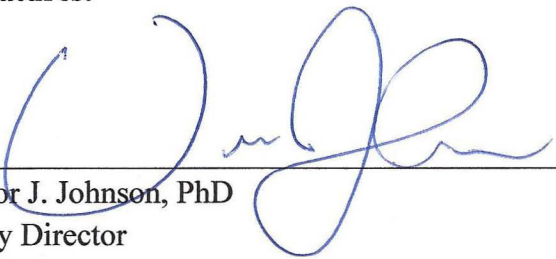
A deviation from the study protocol was discovered after finalizing the study report and closing the study. This amendment is to add the deviation to the study report.

Page 49: ELISA METHOD Section should read;

“1.2 Dilute LPA-Conjugated-BSA coating material to ± 2.5 µg/ml in 0.05 M Carbonate Buffer.”

This amendment is to clarify the concentration of LPA-Conjugated BSA coating material used for validation of the ELISA method.

Signatures:



Victor J. Johnson, PhD
Study Director

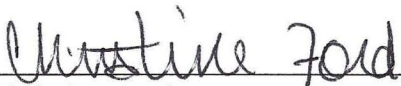
10 Dec 14
Date



Gary R. Burleson, PhD
BRT Management

10 December 2014
Date

QA Review



Christine Ford
BRT QA Manager

10 Dec 14
Date

Report Amendment

Sponsor: Lpath Incorporated

BRT Study No.: BRT 20140629A

Study Title: Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation

Report Amendment No.: 2

Original Report Date: 21 Nov 14

Report Amendment Date: 12 Dec 14

Detailed Description of Amendment

Page 30: CONCLUSIONS Section should read;

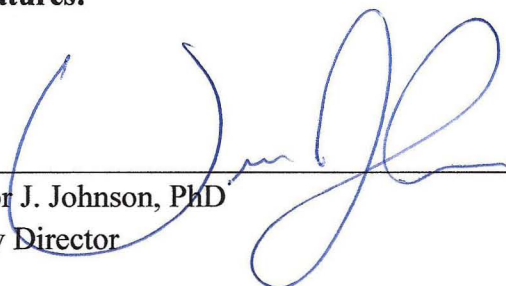
“QC Sample Acceptance Criteria:

- QC1-300 ng/mL; QC2-150 ng/mL; QC3-~~5~~ **10** ng/mL
 - Accuracy of QC samples: 20% of nominal (25% at ~~5~~ **10** ng/mL and 300 ng/mL)
 - Precision of the QCs will be within 20% (25% at ~~5~~ **10** ng/mL and 300 ng/mL)
 - At least 67% of all QC samples will meet these criteria. In addition, at least 50% of the QC samples at each level will meet these criteria.
 - Total error ($|\%RE| + \%CV$) of each QC will be $\leq 30\%$ ($\leq 40\%$ at ~~5~~ **10** ng/mL and 300 ng/mL)”
-

Reason for Amendment

This amendment is required for adjustment of the low QC due to the lower acceptance limit for the 5 ng/mL QC being below the quantifiable range of the calibration curve.

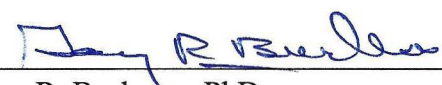
Signatures:



Victor J. Johnson, PhD
Study Director

12 Dec 14

Date

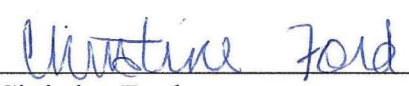


Gary R. Burleson, PhD
BRT Management

12 Dec 2014

Date

QA Review



Christine Ford
BRT QA Manager

12 Dec 14

Date

REPORT ADDENDUM #1

Study Title: Validation of an ELISA Method for the
Quantification of LT3114 in a Citrate Buffer
Formulation

Study Number: BRT 20140629-A

Total Pages 12

Testing Facility: Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560, USA

Report Date: 10 December 2014



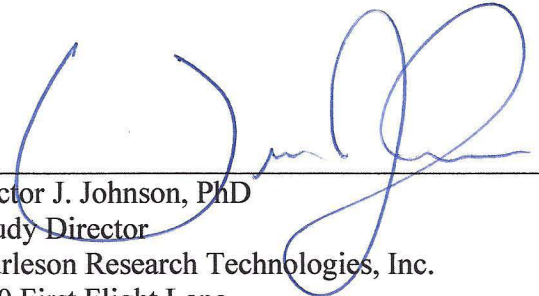
STUDY IDENTIFICATION

Study Title	Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation
Study Number	BRT 20140629-A
Sponsor	Lpath Incorporated 4025 Sorrento Valley Boulevard San Diego, CA 92121
Sponsor Representative	Gary Woodnutt, PhD (Lpath)
Study Monitor	Marlene W. Modi, PhD (Consultant for Lpath) MWM Consulting Group, Inc.
Testing Facility	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Study Director	Victor J. Johnson, PhD (BRT)

COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)
- OECD Principles on Good Laboratory Practice (C[97]186/Final)

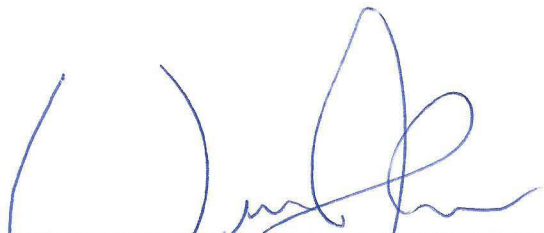


Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
USA

10 Dec 14

Date

APPROVALS



Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.

10 Dec 14

Date

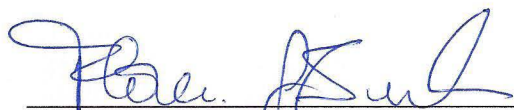


Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

10 December 2014

Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

10 Dec 2014

Date

QUALITY ASSURANCE STATEMENT

Study Number: BRT 20140629-A
Study Title: Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation
Sponsor: Lpath Incorporated
Testing Facility: Burleson Research Technologies, Inc.
Study Director: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to Study Director/Management
Stability Assessment Assay Audit	21 Nov 14	21 Nov 14
Raw Data Audit	05 Dec 14	05 Dec 14
Report Addendum Audit	09 Dec 14	09 Dec 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.


Christine Ford
Quality Assurance Manager


Date

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PURPOSE

The purpose of this report addendum was to extend the long term refrigerated (2-8°C) storage stability assessment for LT3114 in a citrate buffer formulation.

METHODS

ELISA Method

LT3114 in a citrate buffer formulation was stored at 2-8°C at 32.1 mg/mL (as supplied by the Sponsor, manufacture date of June 2014) and 2.0 mg/mL (prepared by BRT on 13 Aug 14). These samples were analyzed for LT3114 concentration according to the methods outlined in the protocol. The ELISA method used for assessment of long term refrigerated (2-8°C) storage stability for LT3114 in a citrate buffer formulation is presented in [Appendix I](#).

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. LT3114 in a citrate buffer formulation, standards, quality control (QC) samples, and dilipidized bovine serum (DBS) assay buffer were added to the coated plates and incubated to allow for capture of the LT3114. A horse radish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Acceptance criteria for demonstrating 2-8°C storage stability of the LT3114 in a citrate buffer formulation was achieved if the analyzed concentration was within 20% of the target concentration.

Data Analysis

Data analysis was performed using the SoftMax®Pro software v 2.2.1 (Spectramax 340). A standard curve was generated by plotting the resulting mean absorbance at 450 nm (OD₄₅₀) vs. concentration and fitting the curve using a 4 parameter logistics (PL) equation. Printouts included absorbance values (raw data), the standard curve(s), the coefficient of variation between replicates, and the interpolated value of the unknown(s). Results were expressed using two decimals in accordance with the decimals used for the calibration curve standards.

Statistical Analysis

No statistical analysis of the data was performed except for calculation of means, standard deviations, %RE (relative error as a measure of accuracy), and %CV (coefficient of variation as a measure of precision), as appropriate. These data calculations were performed using SoftMax®Pro (version 2.2.1 and 5.0.1 as appropriate) and/or Microsoft Excel 2013.

RESULTS

A summary of the assay runs performed to determine stability assessment is provided in [Appendix II](#). Initial assay runs for determination of the stability of LT3114 in a citrate buffer formulation following long term refrigerated (2-8°C) storage indicated that the concentration in the stock being used at BRT was low and did not meet acceptance criteria. An out of specification investigation was subsequently performed and documented in the study records (see [Appendix II](#) for a summary).

In repeat analysis, the mean concentration of LT3114 determined in the 32.1 mg/mL sample was 26.8 mg/mL (83% of target) and in the 2 mg/mL sample was 1.75 mg/mL (88% of target), both of which met the stability acceptance criteria ([Table 1](#)).

Table 1: Concentration of LT3114 in long term refrigerated (2-8°C) storage stability samples.

Assay Date	Target (mg/mL)	Result (mg/mL)	% Accuracy
21 Nov 14	2.0	1.75	88
01 Dec 14	32.1	26.8 ¹	83

¹Mean LT3114 concentration of multiple dilutions (1:80,000, 1:160,000, 1:320,000, 1:640,000, 1:1,280,000) of the 32.1 mg/mL stock analyzed in a single assay run.

CONCLUSION

These results demonstrate that LT3114 in a citrate buffer formulation was stable when stored at 2-8°C for 154 days (32.1 mg/mL sample) and 100 days (2.0 mg/mL sample).

Appendix I: ELISA Method for Quantification of LT3114.

METHOD

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material should be kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material should be stored at 2-8° C for up to 4 months.

2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA I, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 500, 150, and 5 ng/mL. Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 2.5 $\mu\text{g}/\text{ml}$ in 0.05 M Carbonate Buffer (see Deviation in Report Amendment #1).
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 $\mu\text{L}/\text{well}$ coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.
 - 3.2 Dispense 150 $\mu\text{L}/\text{well}$ of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour $\pm$ 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)
 - 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.

- 4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.
8. Stop Solution Addition
 - 8.1 Dispense 100 µl 1.0 M H₂SO₄ solution to each well.
 - 8.2 Measure absorbance at 450 nm on microplate reader.

Appendix II: Summary of assay runs.

Assay Run # (continued from validation)	Assay Run Date	Included in stability assessment (Yes / No)	Comments
19	17 Nov 14	No	Run excluded due to QC samples not meeting acceptance.
20	18 Nov 14	No	LT3114 concentrations were low and did not meet acceptance criteria for stability. QC samples met acceptance criteria. Study Director requested that the assessment be repeated.
21	19 Nov 14	No	LT3114 concentrations were low and did not meet acceptance criteria for stability. QC samples met acceptance criteria. Study records were reviewed and determined that assay was performed according to protocol. Study Director requested that the assessment be repeated for confirmation.
22	21 Nov 14	Yes	- Technician noticed some cloudiness in the DBS assay buffer that was used for the assay runs on 17 Nov 14 and 18 Nov 14. A new vial of DBS from the same lot was used for this and subsequent assay runs. - QC samples met acceptance criteria. 2.0 mg/mL sample met acceptance criteria for stability and data was reported. 32.1 mg/mL sample failed to meet acceptance criteria for stability. Study Monitor was informed. The Study Director suspected contamination of the 32.1 mg/mL stock. The stock was plated but no bacterial or fungal growth was observed, although deterioration of the stock due to repeated use could not be precluded. New stock vials of the 32.1 mg/mL LT3114 were provided at the Sponsors request.
23	25 Nov 14	No	- QC samples met acceptance criteria. 32.1 mg/mL sample result was 25.4 mg/mL (79% of target). - Study Director requested a repeat to confirm result.
24	01 Dec 14	Yes	- All data reported. - QC samples met acceptance criteria. 32.1 mg/mL sample met acceptance criteria for stability. Multiple dilutions of the 32.1 mg/mL stock were analyzed in this run to determine if dilution was a factor in the low result. - Two independent operators also performed the assay on 01 Dec 14. The assay run performed by the second operator was rejected due to the QC samples not meeting the acceptance criteria.

REPORT ADDENDUM #2

Study Title: Validation of an ELISA Method for the
Quantification of LT3114 in a Citrate Buffer
Formulation

Study Number: BRT 20140629-A

Total Pages 11

Testing Facility: Burleson Research Technologies, Inc. (BRT)
120 First Flight Lane
Morrisville, NC 27560, USA

Report Date: 19 December 2014



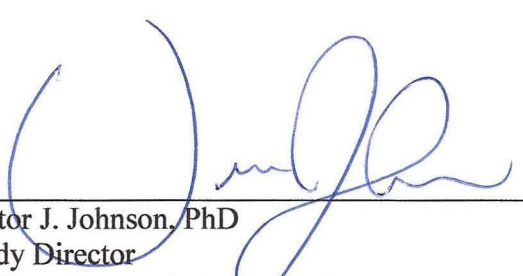
STUDY IDENTIFICATION

Study Title	Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation
Study Number	BRT 20140629-A
Sponsor	Lpath Incorporated 4025 Sorrento Valley Boulevard San Diego, CA 92121
Sponsor Representative	Gary Woodnutt, PhD (Lpath)
Study Monitor	Marlene W. Modi, PhD (Consultant for Lpath) MWM Consulting Group, Inc.
Testing Facility	Burleson Research Technologies, Inc. (BRT) 120 First Flight Lane Morrisville, NC 27560
Study Director	Victor J. Johnson, PhD (BRT)

COMPLIANCE STATEMENT

This study was performed according to the study protocol, applicable BRT SOPs, and in compliance with the following regulations:

- Nonclinical Laboratory Studies Good Laboratory Practice Regulations issued by the U.S. Food and Drug Administration (Title 21 of the Code of Federal Regulations, Part 58)
- OECD Principles on Good Laboratory Practice (C[97]186/Final)

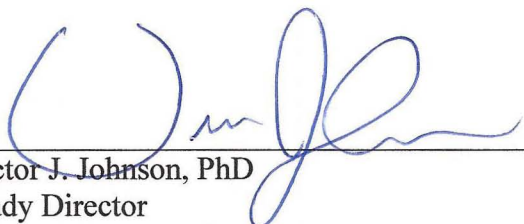


Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.
120 First Flight Lane
Morrisville, NC 27560
USA

19 Dec 14

Date

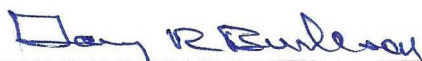
APPROVALS



Victor J. Johnson, PhD
Study Director
Burleson Research Technologies, Inc.

19 Dec 14

Date

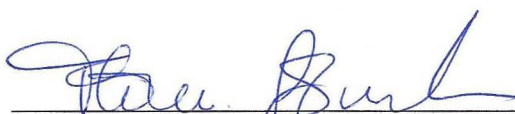


Gary R. Burleson, PhD
President (Management)
Burleson Research Technologies, Inc.

19 Dec 2014

Date

Report Review:



Florence G. Burleson, PhD
Director of Laboratory Operations
Burleson Research Technologies, Inc.

19 Dec 14

Date

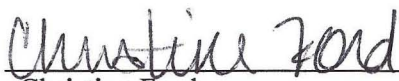
QUALITY ASSURANCE STATEMENT

Study Number: BRT 20140629-A
Study Title: Validation of an ELISA Method for the Quantification of LT3114 in a Citrate Buffer Formulation
Sponsor: Lpath Incorporated
Testing Facility: Burleson Research Technologies, Inc.
Study Director: Victor J. Johnson, PhD

Phase Inspected	Date of Inspection	Date Report Sent to Study Director/Management
Raw Data Audit	19 Dec 14	19 Dec 14
Report Addendum 2 Audit	19 Dec 14	19 Dec 14

The final report and data record audit were performed by the BRT Quality Assurance Unit.

To the best of my knowledge, this final report accurately describes the study methods and procedures used, and the reported results accurately reflect the raw data.


Christine Ford
Quality Assurance Manager


Date

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PURPOSE

The purpose of this report addendum was to extend the long term frozen and refrigerated storage stability assessment for LT3114 in a citrate buffer formulation.

METHODS

ELISA Method

LT3114 in a citrate buffer formulation was stored at 2-8°C at 32.1 mg/mL (as supplied by the Sponsor, manufacture date of June 2014) and 2.0 mg/mL (prepared by BRT on 13 Aug 14). The QC samples prepared on 13 Aug 14 were stored frozen at $\leq -70^{\circ}\text{C}$. These samples were analyzed for LT3114 concentration according to the methods outlined in the protocol. The ELISA method used for assessment of long term frozen and refrigerated storage stability for LT3114 in a citrate buffer formulation is presented in [Appendix I](#).

LT3114 is a human monoclonal IgG antibody specific for lysophosphatidic acid (LPA). The ELISA method for quantification of LT3114 utilized LPA conjugated to BSA as the capture reagent. High protein binding microtiter plates were coated overnight with LPA-BSA. LT3114 in a citrate buffer formulation, standards, quality control (QC) samples, and delipidized bovine serum (DBS) assay buffer were added to the coated plates and incubated to allow for capture of the LT3114. A horseradish peroxidase (HRP)-conjugated secondary antibody specific for human IgG was added and 3,3',5,5'-tetramethylbenzidine (TMB) used as the substrate. Quantification of substrate absorbance was performed using a microplate spectrophotometer using a 450 nm wavelength. Concentrations were extrapolated from the calibration curve and adjusted for dilution factor, if required.

Acceptance criteria for demonstrating 2-8°C storage stability of the LT3114 in a citrate buffer formulation was achieved if the analyzed concentration was within 20% of the target concentration.

Data Analysis

Data analysis was performed using the SoftMax®Pro software v 2.2.1 (Spectramax 340). A standard curve was generated by plotting the resulting mean absorbance at 450 nm (OD_{450}) vs. concentration and fitting the curve using a 4 parameter logistics (PL) equation. Printouts included absorbance values (raw data), the standard curve(s), the coefficient of variation between

replicates, and the interpolated value of the unknown(s). Results were expressed using two decimals in accordance with the decimals used for the calibration curve standards.

Statistical Analysis

No statistical analysis of the data was performed except for calculation of the accuracy of the LT3114 concentration relative to the target concentration. These data calculations were performed using SoftMax®Pro (version 2.2.1 and 5.0.1 as appropriate) and/or Microsoft Excel 2013.

RESULTS

The 150 ng/mL and 5 ng/mL QC samples prepared on 13 Aug 14 were analyzed on 05 Dec 14 resulting LT3114 concentration of 140.8 ng/mL (94% of target) and 4.9 (98% of target), respectively for the 150 ng/mL and 5 ng/mL QC samples, respectively. These analyses met acceptance criteria for accuracy and precision, thereby supporting frozen storage stability of LT3114 in a citrate buffer formulation. The 2.0 mg/mL refrigerated stability sample prepared on 13 Aug 14 was analyzed on 15 Dec 14 and the result was high (4.5 ng/mL, 225% of target) and did not meet acceptance criteria for accuracy (within 20% of target). The paperwork was reviewed and no specific reason could be identified for the high result. The Study Director requested another analysis be performed to include an assessment of the 2.0 mg/mL and the 32.1 mg/mL (as supplied by the Sponsor, manufacture date of June 2014) refrigerated storage stability samples. The assay performed on 16 Dec 14 showed LT3114 concentrations of 2.3 mg/mL (115% of target) and 37.0 mg/mL (115% of target) for the 2.0 mg/mL and 32.1 mg/mL refrigerated stability samples, respectively demonstrating storage stability under these conditions.

CONCLUSION

These results demonstrate that LT3114 in a citrate buffer formulation was stable when stored at $\leq -70^{\circ}\text{C}$ for 114 days, $2-8^{\circ}\text{C}$ for 169 days (32.1 mg/mL sample) and 125 days (2.0 mg/mL sample).

Appendix I: ELISA Method for Quantification of LT3114.

METHOD

Materials and Equipment

- LPA-BSA Conjugated Provided by Lpath
- Human Reference Anti-LPA IgG (Provided by Lpath; LT3114)
- Citrate buffer provided by Lpath
- LT3114 Citrate Formulated Drug Substance (Provided by Sponsor)
- 0.05 M Carbonate Buffer, pH 9.6 (Sigma C3041, or equivalent)
- Phosphate Buffered Saline (PBS, 1X) (Gibco 10010, or equivalent)
- Polyoxyethylene sorbitan monolaurate (Tween-20) (Sigma-Aldrich – Cat # P-1379 or equivalent)
- Wash solution (PBS, 0.1% Tween-20)
- Blocking Solution (PBS, 0.1% Tween-20, 1% BSA)
- Delipidized Bovine serum (DBS), assay diluent (Biocell cat 1231-00 or equivalent)
- Albumin: Bovine Serum, Fraction V, Fatty Acid-Free (Cat # 126575, CALBIOCHEM phone # 858-450-5558)
- Peroxidase Conjugated anti-human IgG (Jackson 709-035-149)
- TMB substrate (3,3',5,5'-Tetramethylbenzidine) liquids substrate system for ELISA (Sigma T0440, or equivalent)
- 1M H₂SO₄ (Mallinkrodt H381, or equivalent)
- Microtiter plates for ELISA: 96 well flat bottom, high binding polystyrene (Greiner 655001, or equivalent)
- Cluster tubes (VWR 29442-610, or equivalent)
- Plate Sealers
- Disposable polypropylene containers: 15 mL, 50 mL, 100 mL
- Weighing Boats
- Pipette tips for pipettors
- Sterile 0.2 µm filter units (VWR 87006-066, or equivalent)
- Disposable serological pipettes: 5 mL, 10 mL, 25 mL, 50 mL, 100 mL
- Micropipettors
- Microtiter plate washer
- Bench top vortex mixer
- ELISA microtiter plate reader

Solutions and Storage

1. LPA-Conjugated-BSA (provided by the Sponsor)
 - 1.1 LPA-Conjugated-BSA coating material was kept frozen until use.
 - 1.2 After dilution, LPA-Conjugated-BSA coating material was stored at 2-8° C for up to 4 months.

2. Peroxidase conjugated-anti-human IgG secondary antibody
 - 2.1 Reconstitute following vendor instructions, Dilute in glycerol (50%).
 - 2.2 Store at -20° C for up to 1 year.
3. Carbonate Buffer (0.05 M Carbonate Buffer pH 9.6)
 - 3.1 Add 1 tablet per 100 mL of deionized water, mix thoroughly.
 - 3.2 Store at 2-8°C for up to 3 months.
4. Blocking Solution 1xPBS, 1% BSA l, 0.1% Tween 20
 - 4.1 Dissolve 10 g BSA and 1 mL Tween-20 in 999 mL PBS (1X).
 - 4.2 Mix thoroughly and sterile filter.
 - 4.3 Store at 2-8°C for up to 1 month.

Preparation of Quality Control Samples

Prepare QC samples in citrate buffer using Human Reference Anti-LPA IgG LT3114 at 300 (Report Amendment 1), 150, 5 ng/mL (05 Dec 14 assay run) and 10 mg/mL (15 Dec 14 and 16 Dec 14 assay runs; Report Amendment 2). Aliquot and store at $\leq -70^{\circ}\text{C}$.

ELISA Method

1. Coating Plates
 - 1.1 Retrieve the appropriate frozen aliquot of LPA-Conjugated-BSA that was stored at $\leq -70^{\circ}\text{C}$. Thaw the aliquot at room temperature.
 - 1.2 Dilute LPA-Conjugated-BSA coating material to 2.5 $\mu\text{g}/\text{ml}$ in 0.05 M Carbonate Buffer (see Deviation in Report Amendment #1).
 - 1.3 Coat a 96-well high-binding microtiter plate(s) with 100 $\mu\text{L}/\text{well}$ coating solution.
 - 1.4 Cover the plate(s) with plate sealers and incubate plate overnight at 2-8°C.
2. Bring all materials and plate(s) to room temperature before starting assay (except TMB).
3. Blocking Plates
 - 3.1 Wash coated plate(s) 4 times with wash solution.
 - 3.2 Dispense 150 $\mu\text{L}/\text{well}$ of blocking solution to each well and cover the plate(s) with plate sealers.
 - 3.3 Incubate plate(s) at 37°C for 1 hour $\pm$ 5 minutes.
4. Prepare Samples/Reference Controls/Blanks while blocking plate(s)
 - 4.1 Retrieve samples from storage and allow them to thaw at room temperature. (Note: Minimize freeze thaw cycles)

- 4.2 Prepare unknown samples by diluting in DBS to appropriate dilutions.
 - 4.3 Prepare Human Reference Anti-LPA IgG 12 point standard curve by diluting in DBS to achieve desired range (3.9 – 500 ng/ml).
 - 4.4 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 4.5 Add samples (controls, calibrators and unknowns) at 100 µL/well.
5. Add samples to appropriate wells
 - 5.1 Dispense 100 µl of appropriate sample to designated wells in duplicate. Add 100 µl of DBS to assay blank wells.
 - 5.2 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
6. Add HRP-goat-anti-human secondary antibody solution.
 - 6.1 Wash plate(s) 4 times (300µl/well) with wash solution.
 - 6.2 Create the working dilution of 1:40,000 of HRP-goat-anti-human secondary antibody
 - 6.2.1 Dilute HRP conjugated anti-human IgG with blocking solution to 1:20 from the stock (50% glycerol), creating a 1:40 dilution.
 - 6.2.2 Using 1:40 dilution, further dilute to a 1:1,000 final dilution in blocking solution.
 - 6.3 Dispense 100 µl of diluted HRP-goat-anti-human secondary antibody to each well.
 - 6.4 Cover the plate(s) with plate sealers and incubate plate at 37°C for 1 hour ± 5 minutes.
7. Substrate Addition
 - 7.1 Wash coated plate(s) 4 times with wash solution.
 - 7.2 Dispense 100 µl **COLD** TMB solution to each well.
 - 7.3 Cover the plate(s) with plate sealers and incubate for 5-30 minutes protected from light.
8. Stop Solution Addition
 - 8.1 Dispense 100 µl 1.0 M H₂SO₄ solution to each well.
 - 8.2 Measure absorbance at 450 nm on microplate reader.